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Review

Mechanically modeling chemical mechanical planarization from wafer to molecular scales: a review and discussion of future direction

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ABSTRACT

The chemical mechanical planarization (CMP) process plays an increasingly important role in integrated circuit (IC) manufacturing. To understand the mechanism broadly, we review different scales of models describing the removal phenomenon and predicting the material removal rate (MRR). At the wafer scale, there is the empirical Preston equation and its extensions incorporating different mechanical and chemical factors. Toward feature and particle scales, we introduce several models postulating the material removal mechanism and their experimental verification. Finally, we mention several mechanical models on the molecular scale and propose that at the moment they lack generality and do not give us further useable, quantifiable insight into the CMP phenomenon. We conclude with how we may, and should, achieve finding this mechanical model on a molecular scale.

KEYWORDS: CMP, Greenwood–Williamson model, mixed lubrication, molecular scale

1 INTRODUCTION

Chemical mechanical planarization (CMP) is a crucial manufacturing technique in the manufacturing of integrated circuits (ICs), which can be fundamentally understood as a layers-upon-layers building process. From the front end of forming transistors with its isolating trenches to the back-end interconnects enabling the logic circuits, planarization is required every step of the way so the whole "building" can be structurally stable, contact spots can be properly aligned and appropriate barrier can be laid down so different materials will not mutually interfere. Two trends in the development of IC further accentuate the importance of the role CMP plays, one being the continued shrinking of the transistor node size requiring ever so extreme planarization up to the Angstrom level, another being the advent of 3D IC requiring more layers to be stacked. To this day, CMP is still the technology being relied upon to fulfill these ever-so-demanding needs. In the CMP process, the wafer is held by a carrier with the intended polished side facing down, applying load and pressing against a large polishing pad as shown in Fig 1. Both the wafer and the pad are rotating along their respective axis. While rotating, chemical slurry with nanoscale abrasive particles is delivered onto the pad surface, creating a lubricating thin film between the wafer and the pad. This is where the mechanical and chemical aspects synergize. The chemical slurry reacts with the wafer, softening the surface, while the abrasives trapped between the wafer and pad mechanically remove the reacted layer.

Since the introduction of the method from IBM lab in 1986 [1], this concept has been utilized in a wide range of materials, from pure silicon, silicon oxide and nitride [2], III–V material [3], ultra low-k porous material [4], metals like tungsten and copper, and more recently crystalline material like silicon carbide and sapphire [5]. Planarization of each material requires its appropriate chemicals, abrasive type and corresponding pad variations. Commercially, the optimization of CMP parameters has been evolving at a rapid pace, while at the same time, academically, researchers have been constructing models to understand the planarization mechanisms, trying to predict the removal rate and the resulting surface roughness. In this review article, we aim to organize different proposed models in the order of macro to micro scales, for each model has its own analyzing perspective, with some factors and mechanisms considered and some neglected. From the wafer scale con-

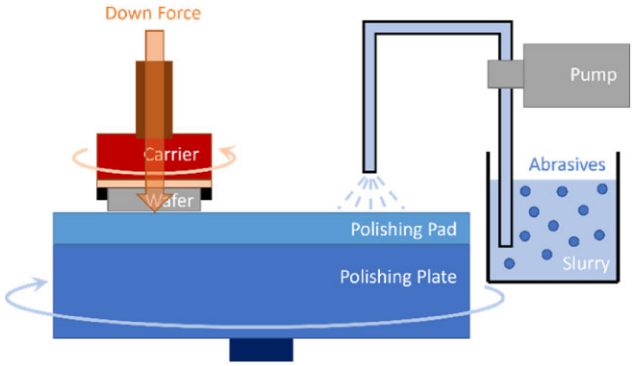


Figure 1 CMP apparatus schematics.

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sidering the pressure and velocity, to the feature scale of how surface configuration affects the removal rate, further down to the particle scale regarding the contact mechanism and down to the molecular scale of a single molecule being reacted and dissolved, we hope this overall view can give the readers a brief but adequate summary of this research field. Then toward the latter part of the article, we illustrate our current research dilemma and propose a future direction.

2 WAFER SCALE

With CMP sharing its roots in the polishing sphere, a simplified empirical model originally used for glass polishing, the Preston equation, is still being utilized to this day [6–8].

$$\mathrm{MRR} = K_p \times P \times V,$$

where K_p is the Preston coefficient. It states that the material removal rate (MRR) is directly proportional to the wafer contact pressure (P) times the relative velocity (V).

2.1 Preston equation and its extension

A simple linear combination of pressure and velocity does not quite accurately capture the complex phenomena of CMP. Tseng and Wang [9] considered contact mechanics and proposed a non-linear equation:

$$\mathrm{MRR} = K_p \times P^{\frac{5}{6}} \times V^{\frac{1}{2}},$$

and verified the result experimentally, showing a better fit compared to the original Preston equation [10]. Fu *et al.* [11] considered different particle shapes and pad material and proposed respective removal rate equations . Sorooshian *et al.* [12] incorporated the reacting temperature and amended the removal rate equation with the Arrhenius relation.

2.2 Localized Preston equation

The above model tries to fit the removal rate by modifying the Preston equation; however, they all ignore the factor of within wafer non-uniformity, where throughout the wafer surface, there is a spatially varying distribution of different velocity and stress [13], resulting in different removal rates at different coordinates. Castillo-Mejia and Beaudoin [14] proposed a localized Preston equation

$$\mathrm{MRR}(x, y) = K_p \times P(x, y) \times V(x, y),$$

and kinetically analyzed the velocity distribution. As for pressure, multiple researchers have analyzed the stress distribution under different circumstances. Che *et al.* [15] measured the stress distribution and investigated how it enhances the mass removal process. Chen [16] and Lee and Jeong [17, 18] used finite element analysis to determine the stress distribution and verified it with experimental results. At the edge of the wafer, the retaining mechanism can greatly affect the stress concentration. Wang and Lu [19] proposed a multi-zone pressure-retaining mechanism to make the pressure distribution more uniform.

3 FEATURE TO PARTICLE SCALES

The above models expanded on the Preston equation and remedied its empirical nature by incorporating theoretical elements into the equation by altering the linear combination of pressure and velocity or expanding on the Preston coefficient with material properties or chemical reaction factors. It achieved great predictability, but it still came from an intuitive assumption that the removal rate is related to pressure and velocity. We can, or in pursuit of accuracy, we should, go beyond the assumption and try to construct the model from the actual mass removal mechanism.

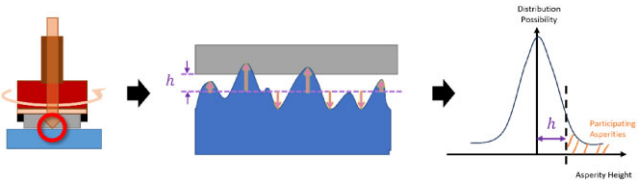


Figure 2 The wafer–pad true contact area illustration (adapted from [13])— pad is treated as a non-uniform material with a normal distribution profile based on an average height of zero, where only height larger than h will make contact with the polishing substrate.

3.1 Greenwood–Williamson model and contact mechanic

Despite most illustrations of the whole CMP process presenting the polishing pad as a flat surface, under microscopic view, it is indeed wavy, with highs and lows that researchers like to attribute the term "asperity" to. The Greenwood–Williamson model generalizes that these highs and lows can be seen as a normal distribution, and only those with sufficiently high asperity can participate in the contact [20]. Recent research with real optical measurement shows that the contact spot radius does not strictly follow a normal distribution [21], but mathematically the measured distribution can be simply substituted into the normal distribution in the Greenwood–Williamson model without sacrificing the integrity of the theory.

Then comes the task of figuring out how many particles are trapped in those contact spots. Some assumed a slurry with uniform abrasive size distribution and uniform concentration [22], while others assumed that particle size is also normally distributed, and only those large enough can participate in material removal [8,23] as illustrated in Fig 2. Next is the step of calculating how deep each particle can indent into the wafer. By assuming a spherical particle sandwiched between the wafer and pad, with the particle–wafer contact being plastic and the particle–asperity contact being the elastic [8, 24, 25], and aided by the fact that the top and bottom forces should be balanced, the resulting indentation depth, and hence the final MRR, can be theoretically deduced [26]. Here, the final MRR formula not only covers pressure and velocity, but also incorporates particle size, particle and asperity size distribution, particle shape [27], and all the participating sides' mechanical properties. Lin *et al.* [28] improved the model by measuring the material properties before and after the slurry reaction, and Akbar and Ertunc [29] attributed the change in the reacted wafer's properties to water diffusion and modeled accordingly. Despite only pad asperities participating in the abrasion, the bottom "bulk" part can still affect the whole deformation of the pad, indirectly affecting the material removal process. Fan [30] and Jeong *et al.* [31] have constructed models incorporating bulk deformation.

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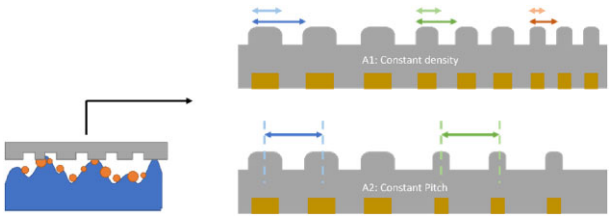


Figure 3 The non-uniform wafer surface with pattern density illustration—the density is defined by length of feature over base+feature, while the pitch is defined as distance between the mid points of the features.

3.2 Pattern density

The above section dealt with the issue of the polishing pad being a rough surface. Although not so relatively comparable length scale-wise, the wafer can also be non-uniform. This normally occurs due to the underlying transistors being of different densities for other areas; hence the deposited material will have different densities and pitches, of which definition is illustrated in Fig 3 [32, 33].

With the fact of the pad being deformable, the bottom step can start being polished despite the top step height not being removed. This circumstance can be dealt with by assuming an intermediate height where bottom removal commences [34], making a geometric assumption of the deformed pad [35], or calculating an "effective pattern density" to transform the discrete pattern density into a continuous curvature surface [36, 37].

3.3 Hydrodynamic and mixed lubrication model

Researchers have also tried to approach the CMP process with the thin film formed by a chemical slurry between the wafer and the polishing pad. By assuming Newtonian, steady-state fluid in between, with the top and bottom being flat plates with known boundary velocity from rotational kinematics, one can solve the simplified Navier–Stokes equation and get the stress distribution on the wafer [38–40]. Jin *et al.* [41] aimed to make the model closer to a real CMP phenomenon, hence considering the continuously deforming asperity, and modified the control equation and boundary conditions accordingly, then tried to get the result iteratively. Although a new perspective, completely ignoring the abrasives and chemicals affecting the liquid and boundary is far from ideal. In traditional tribology, there is an empirical "Stribeck curve," relating slurry viscosity, pressure, and pressure to the friction coefficient experienced by the polishing surface as shown in Fig 4.

Han *et al.* [42, 43] tried to construct the Stribeck curve using CMP slurry and discovered that different polishing scenarios, with different surface materials and slurry components, contributed to different curves. Crawford *et al.* [44] also discovered a major change in viscosity during the material removal process attributed to particle agglomeration , making the previous Newtonian fluid assumption void. Higgs *et al.* [45, 46] tried to solve hydrodynamics, contact mechanics, and tribology wear all together with a multi-physics simulation called particle-augmented mixed lubrication, but such a model required complete morphology of both the pad and the wafer, the exact material being

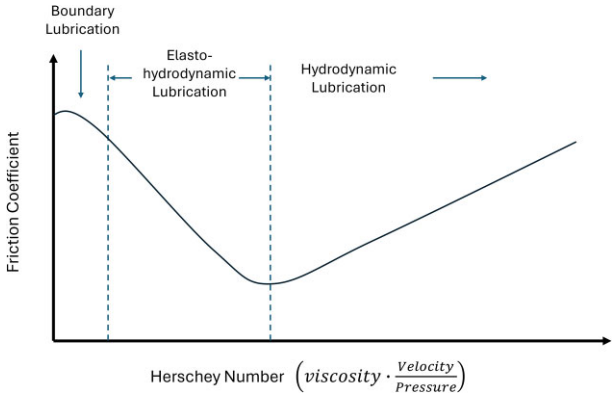


Figure 4 Stribeck curve.

polished, the exact abrasive and slurry used, which in the author's view sacrifices the generality required for a mechanics model.

4 MOLECULAR SCALE, VIRTUAL METROLOGY AND JOURNEY AHEAD

The lack of generality is also a sacrifice we face when we move to the molecular scale model. Molecular dynamics (MD) is a sphere of research that has garnered much interest. Nguyen and Fang [47, 48]have conducted several studies simulating how the abrasives' motion state, such as rolling or impact angle, affects the surface removal phenomenon. However, the type of abrasive and wafer surface material all have to be specific, and the results can only be presented with numerous images, without proper quantification of the removal rate. Si *et al.* [49] also utilized the MD method to demonstrate the monoatomic layer removal characteristic of CMP. However, its application is also restricted to a silicon wafer and a single silica abrasive, let alone ignoring the hydrodynamic and chemical effects.

Another sphere of research, like automatic virtual metrology, has also been explored by many researchers. Originating from applications like endpoint detection and error detection [50, 51], sensors embedded in the CMP apparatus or later attached externally are used to read out the manufacturing parameters' readings and predict the final polished performance [52, 53]. This method, although commercially useful, reduces the removal mechanism to a black box, is not explainable and barely fits the definition of a phenomenon-explaining model.

5 CONCLUSION

We have gone through the wafer scale empirical equations and their further theoretical extensions, to feature and particle scale models where the removal mechanism can be postulated, and observed, and where the result can be characterized with removal rate. However, extending to mixed lubrication, and further to the molecular scale, without resulting in pure chemistry, it is difficult to establish a good generality model.

A point that is worth reiterating is that it is not the smaller the scale the better, it all depends on the need and problems faced by each researcher. It is just that the lack of molecular scale investigation makes the field stagnate that we cannot reach the next frontier of surface smoothness. The larger scale of such

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models still provides valuable information and is required of innovation as well. This also leads to another point, that a model can reach across different scales, from wafer to molecular, all based on researchers' consideration. The categorization in this review is valuable in the way that it puts different models into perspective of novice researchers, and points toward a future direction, not being a textbook–ironclad definitive principle.

The author strongly believed that this is not a spectrum where you must balance scale and generality and make sacrifices. Perhaps the answer lies in further material science, or maybe virtual metrology can tell us which parameter to tune while performing CMP, hence discovering new parameters' relation and postulating new models in the process. The author also believes, with the ever-so-extreme surface roughness demand of the IC industry, that mechanically understanding the CMP process on a molecular level is a must to break the barrier.

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AUTHOR CONTRIBUTIONS

Both authors contributed equally to the writing.

CONFLICT OF INTEREST

No competing interest is declared.

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